

EE4410 CYBER SECURITY OF POWER GRIDS

Assignment 5: Anomaly Detection and Time Series Classification

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Zubair Al-Zubi (5342007)

[Link to Google Colab Notebook](#)

This lab investigates two approaches for detecting cyber attacks in power grid operational technology (OT) networks: Traffic Data & Throughput Graph (TDG) analysis and Time Series Classification (TSC) using machine learning. The analysis covers three scenarios (X, Y, Z) representing different network conditions in a digital substation environment.

1 Network Traffic Analysis Using Throughput and TDG

1.1 Classification of Scenarios

The throughput plots in **Figure 1** show clear differences across the three PCAP files. PCAP X and Z exhibit stable, periodic throughput values, consistent with normal operational traffic. In contrast, PCAP Y contains a distinct burst with a peak above 60,000 Bytes/s, which strongly indicates cyber attack behavior such as flooding or injection.



Figure 1: PCAP throughput subplots

1.2 Adversary and Target Identification

The packet count distribution across source IPs is shown in **Figure 2** for each scenario.

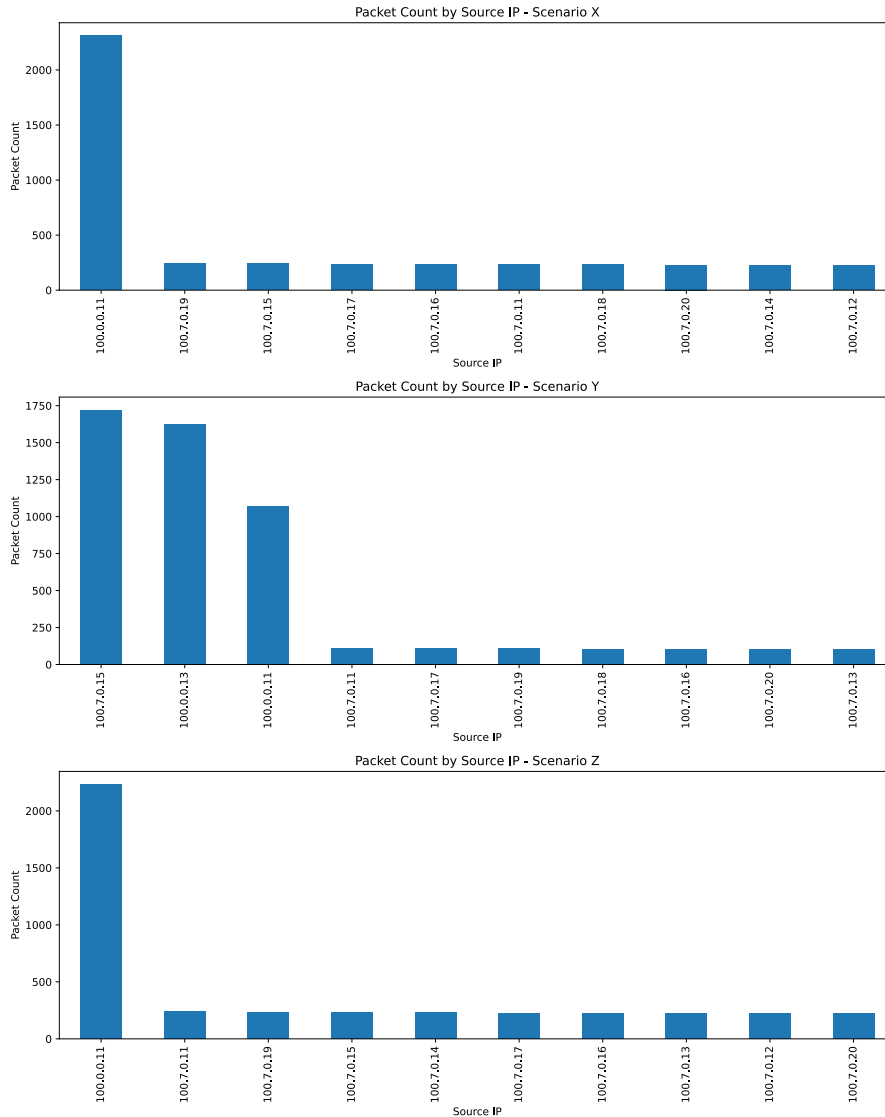


Figure 2: Packet count by source IP for scenarios X, Y, and Z

In Scenario X, IP address 100.0.0.11 is likely the attacker because it sends far more packets and data than any other IP address. The other IP addresses, mostly in the 100.7.x.x range, receive a consistently low amount of traffic, suggesting they are targets of a widespread scan or probe.

In Scenario Y, IP addresses 100.7.0.15, 100.0.0.13, and 100.0.0.11 send high amounts of traffic. These IPs are likely coordinated attackers participating in a distributed attack. The remaining IP addresses receive low and similar amounts of traffic, implying they are targets, possibly of a distributed denial-of-service (DDoS) attack.

In Scenario Z, IP address 100.0.0.11 again sends the most packets and data, strongly suggesting it is the attacker. It is likely scanning or probing the other IPs in the 100.7.x.x range, which are the targets.

The concentration of traffic volume by source is shown in **Figure 3**.

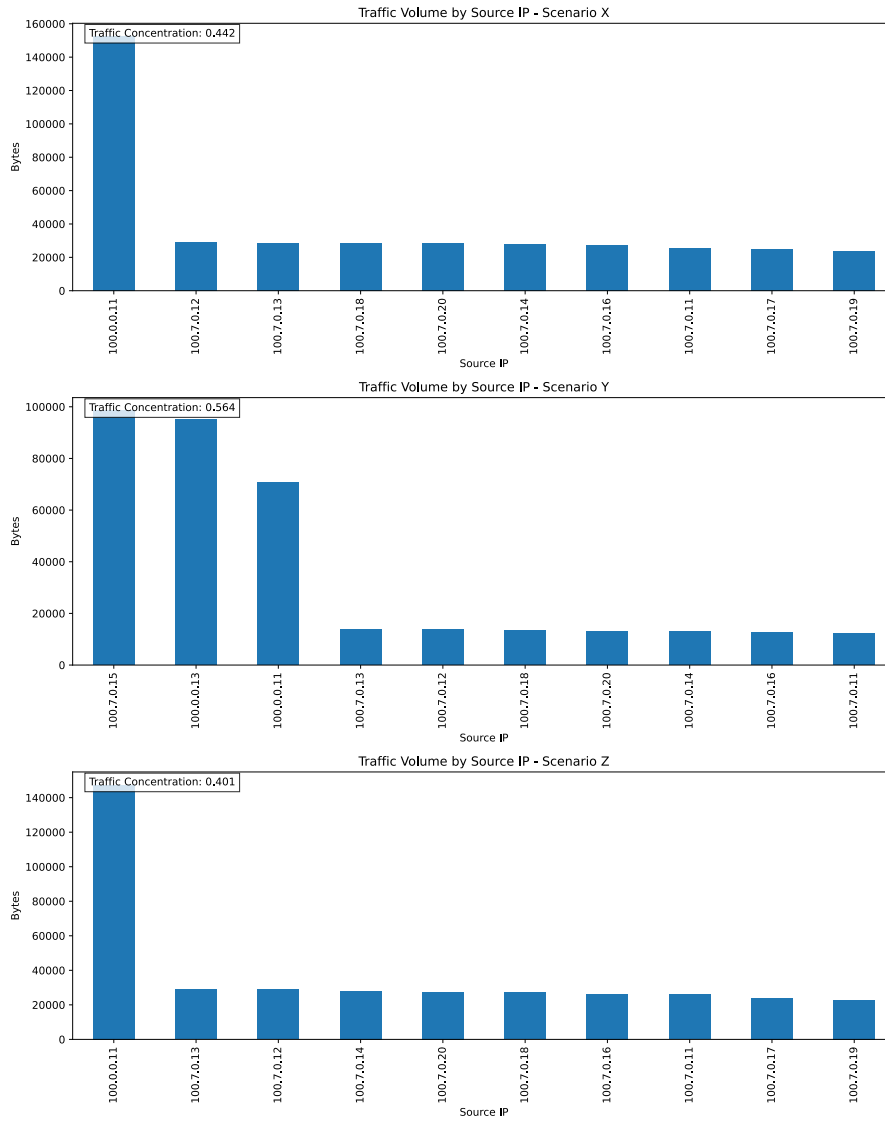


Figure 3: Traffic volume concentration by source IP for scenarios X, Y, and Z

2 Time Series Classification using Machine Learning

A convolutional neural network (CNN) was trained on node A data from `tsc_training_data.csv` using label `A_L`. The model used a sequence length of 4 and classified traffic as normal (0) or attack (1).

The model was applied to `test_X.csv`, `test_Y.csv`, and `test_Z.csv`. Results are shown in Figures [Figure 4](#), [Figure 5](#), and [Figure 6](#).

All traffic in Test X and Test Y was classified as normal. In Test Z, one sample was flagged as an attack, suggesting possible low-level malicious activity.

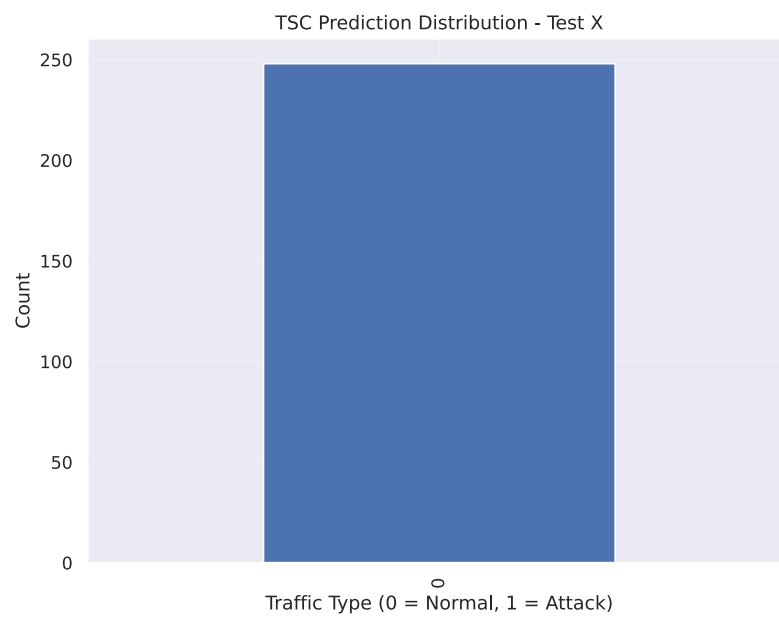


Figure 4: TSC prediction distribution for Test X

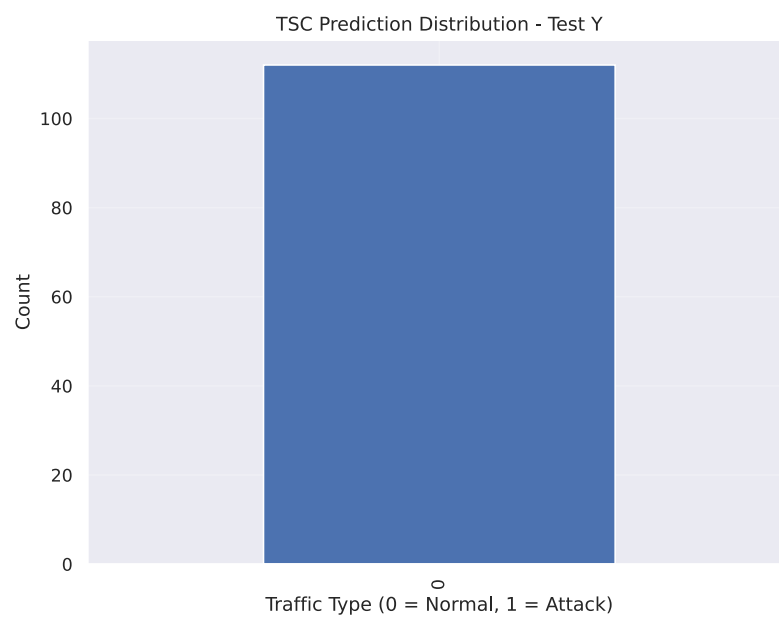


Figure 5: TSC prediction distribution for Test Y

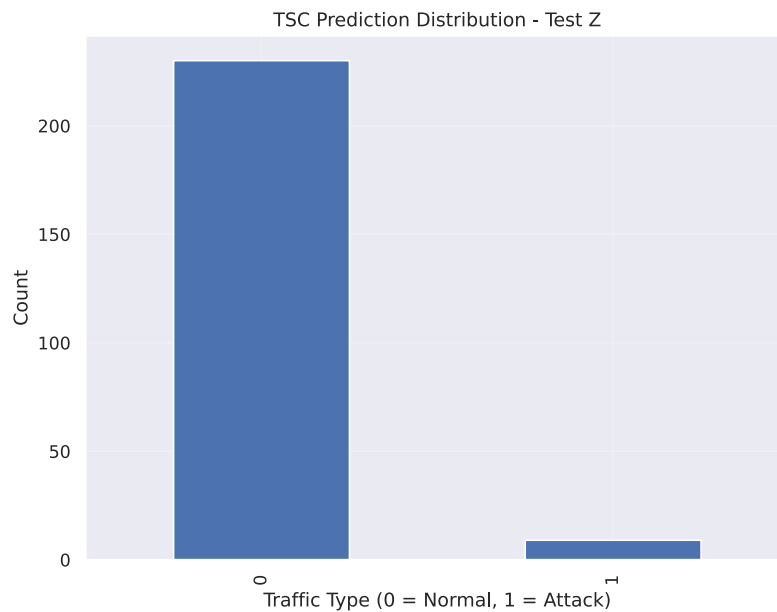


Figure 6: TSC prediction distribution for Test Z

3 Comparative Analysis

This section compares the two methods used in this lab: Traffic Data & Throughput Graph (TDG) analysis and Time Series Classification (TSC).

3.1 Method Comparison

TDG detects anomalies by analyzing throughput patterns and traffic volumes across IPs. It works well for identifying obvious attacks and mapping attacker-target relationships.

TSC uses a trained machine learning model to classify time series data as normal or attack. It is better at catching subtle or low-volume anomalies that may not show up in basic traffic stats.

3.2 Advantages and Drawbacks

TDG

- Simple and easy to implement
- Effective for obvious, high-volume attacks
- Requires manual effort and can be time-consuming
- Less accurate for subtle or time-based attacks

TSC (Machine Learning)

- Automated and consistent
- Detects less obvious patterns
- Needs labeled training data
- May miss new or unusual attack types

3.3 Suggestions for OT Network Monitoring

- Use TDG for fast checks and initial assessment
- Use Time Series Clustering (TSC) to find anomalies based on patterns.
- Update machine learning (ML) models frequently with the latest information.
- Use both TSC and ML together for stronger anomaly detection.